

3.3.3

AI25BTECH11019 - MENAVATH SAI SANJANA

Question:

A line l passes through point $(-1, 3, -2)$ and is perpendicular to both the lines $\frac{x-1}{1} = \frac{y-1}{2} = \frac{z+1}{3}$ and $\frac{x+2}{-3} = \frac{y}{2} = \frac{z}{5}$. Find the vector equation of the line l and its distance from the origin.

Solution:

The two given direction vectors are

$$\mathbf{d}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \mathbf{d}_2 = \begin{pmatrix} -3 \\ 2 \\ 5 \end{pmatrix}. \quad (0.1)$$

Let the required direction vector be $\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$. It satisfies

$$\mathbf{d}_1^\top \mathbf{u} = 0, \quad (0.2)$$

$$\mathbf{d}_2^\top \mathbf{u} = 0. \quad (0.3)$$

Substituting values:

$$u_1 + 2u_2 + 3u_3 = 0, \quad (0.4)$$

$$-3u_1 + 2u_2 + 5u_3 = 0. \quad (0.5)$$

Eliminating u_1 :

$$8u_2 + 14u_3 = 0 \Rightarrow u_2 = -\frac{7}{4}u_3. \quad (0.6)$$

Substitute in the first equation:

$$u_1 + 2\left(-\frac{7}{4}u_3\right) + 3u_3 = 0 \Rightarrow u_1 = \frac{1}{2}u_3. \quad (0.7)$$

Choosing $u_3 = 4$ gives integer values:

$$\mathbf{u} = \begin{pmatrix} 2 \\ -7 \\ 4 \end{pmatrix}. \quad (0.8)$$

Hence the line through $(-1, 3, -2)$ is

$$\mathbf{r} = \begin{pmatrix} -1 \\ 3 \\ -2 \end{pmatrix} + t \begin{pmatrix} 2 \\ -7 \\ 4 \end{pmatrix}, \quad t \in \mathbb{R}. \quad (0.9)$$

For the closest point to the origin, the parameter is

$$t = -\frac{\mathbf{P} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} = -\frac{(-1)(2) + 3(-7) + (-2)(4)}{2^2 + (-7)^2 + 4^2} = \frac{31}{69}. \quad (0.10)$$

So the foot of the perpendicular is

$$\mathbf{Q} = \mathbf{P} + t\mathbf{u} = \begin{pmatrix} -\frac{7}{69} \\ -\frac{10}{69} \\ -\frac{14}{69} \end{pmatrix}. \quad (0.11)$$

Distance from the origin:

$$D = \|\mathbf{Q}\| = \sqrt{\frac{5}{69}}. \quad (0.12)$$

Conclusion: The required line is

$$\mathbf{r} = \begin{pmatrix} -1 \\ 3 \\ -2 \end{pmatrix} + t \begin{pmatrix} 2 \\ -7 \\ 4 \end{pmatrix}, \quad t \in \mathbb{R}, \quad (0.13)$$

and its distance from the origin is

$$D = \sqrt{\frac{5}{69}}. \quad (0.14)$$

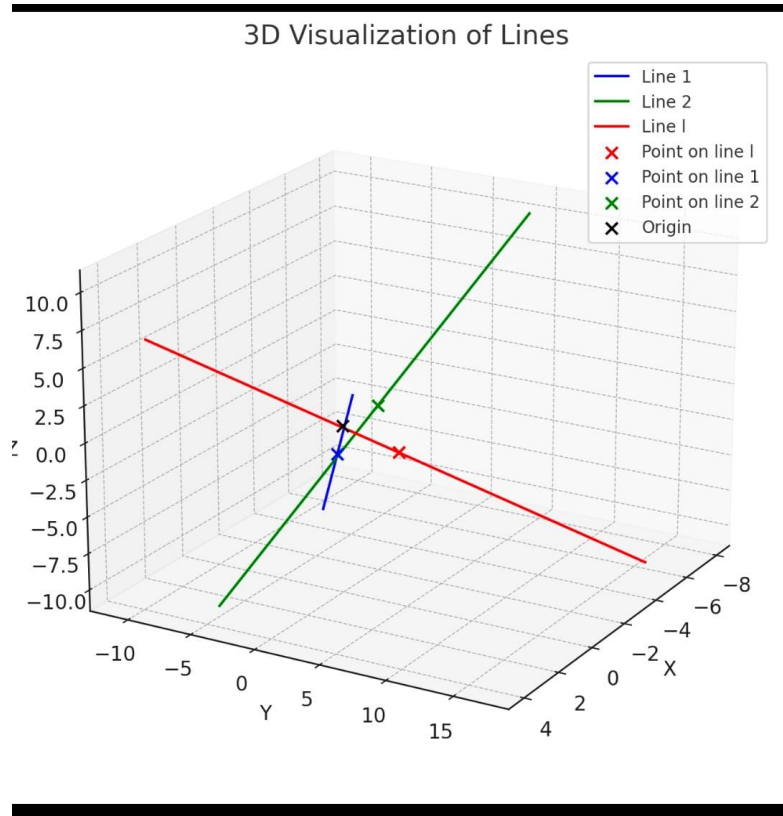


Fig. 0.1